

DATA SCIENCE CAPSTONE SHOWCASE

Predicting Contraceptive Use Among Kenyan Women

A Machine Learning & Deep Learning Approach to
Targeting Family Planning Interventions

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Exploratory Data Analysis

Understanding the demographic, socioeconomic, and geographic landscape

Project Overview & Target Distribution

The Core Problem

Kenya's mCPR is ~57% nationally, hiding a 56-percentage-point gap between counties (Mandera 1% vs Embu 57%).

Goal: predict contraceptive method type to help health planners target resources where non-use is highest.

Target Variable (4-Class)

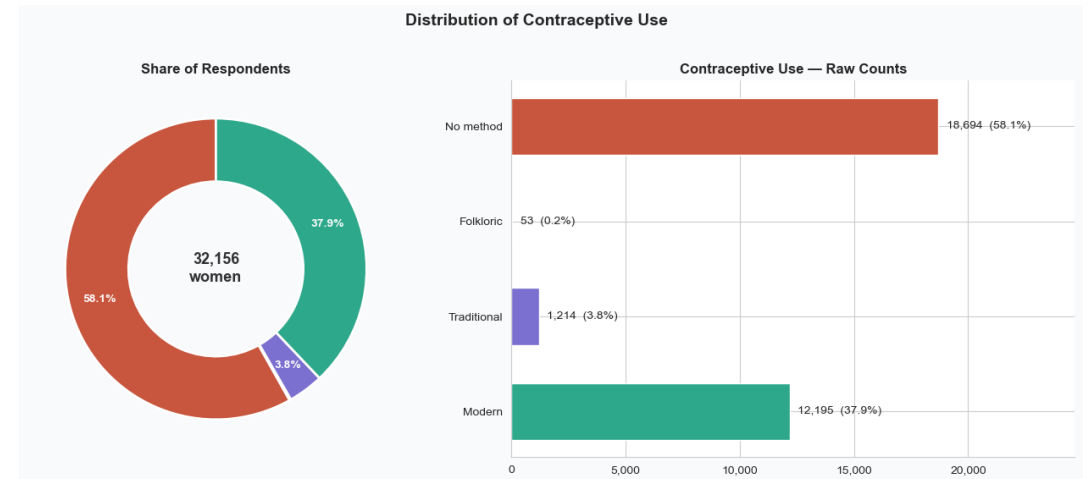
No method: 58.1% (18,694 women)

Modern: 37.9% (12,195)

Traditional: 3.8% (1,214)

Folkloric: 0.2% (53 women only)

The 53-row Folkloric class is the single most important structural constraint in the dataset.



Demographics: Age Distribution

Age Profile

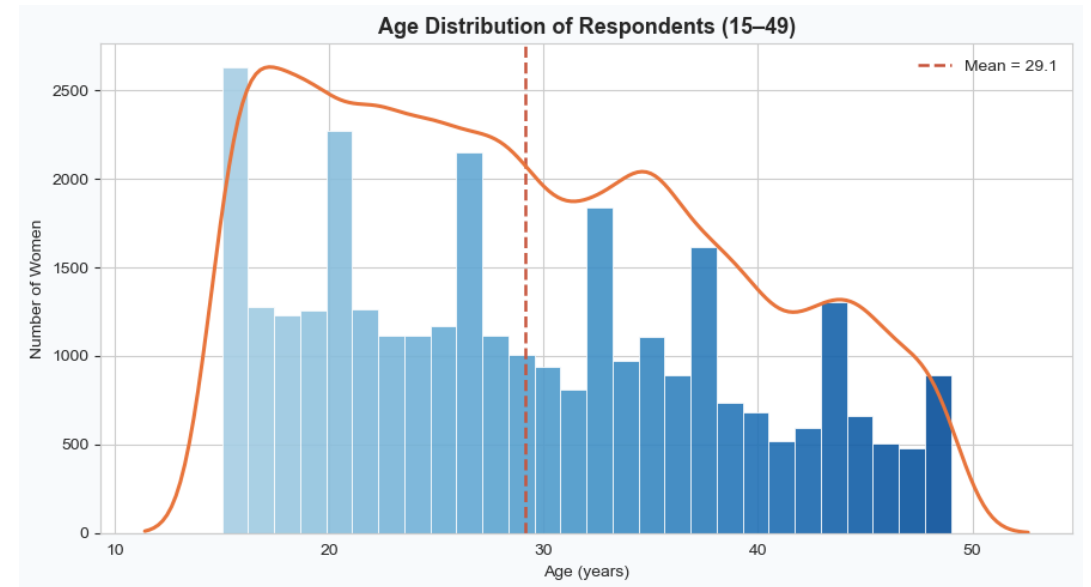
Mean age 29.1, median 28. Survey covers women aged 15–49 across all 47 counties.

Numbers concentrate in the twenties and thin toward 49.

Numeric Summary

Household size and living children are heavily right-skewed.

```
print('Rows and columns:', df.shape) # (32156, 20)
df.describe().T
```



Demographics: Education Level

Education Level Distribution

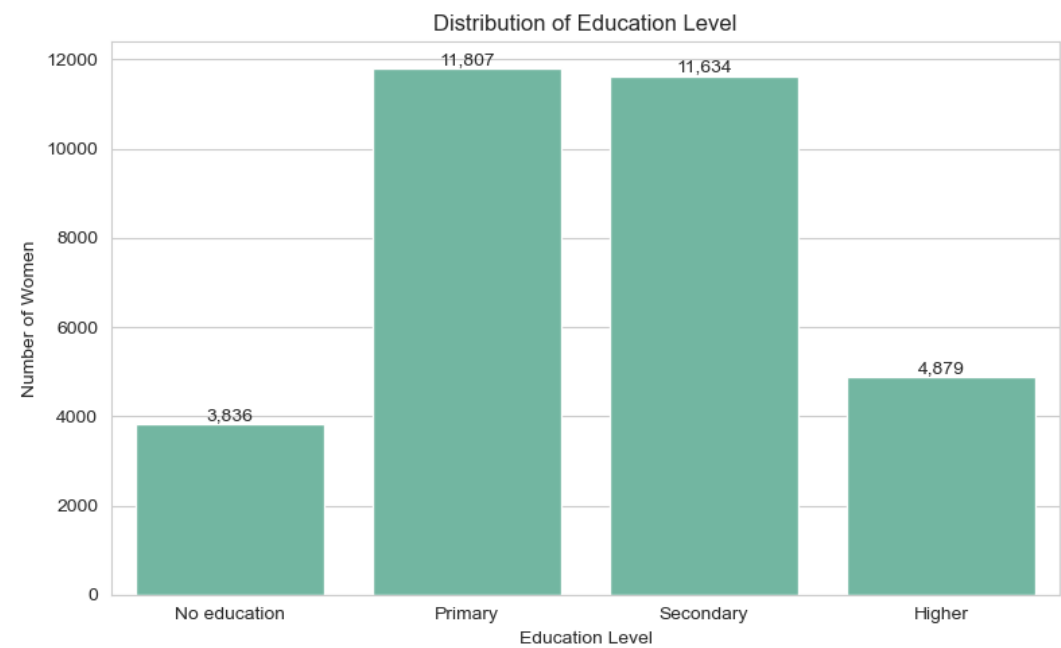
Primary: 36.7%

Secondary: 36.2%

Higher: 15.2%

No education: 11.9%

Since education influences health awareness and access to reproductive health information, this variable is expected to play an important role.



Socioeconomics: Wealth Index & Residence

Wealth Index

Quintiles fairly balanced (18–22% each): Richer 22.3%, Poorest 22.0%, Middle 19.7%, Richest 18.1%, Poorer 17.9%.

Residence Type

Rural: 61.5%

Urban: 38.5%

The rural majority is the primary focus for policy interventions.



Bivariate: Education as a Driver of Uptake

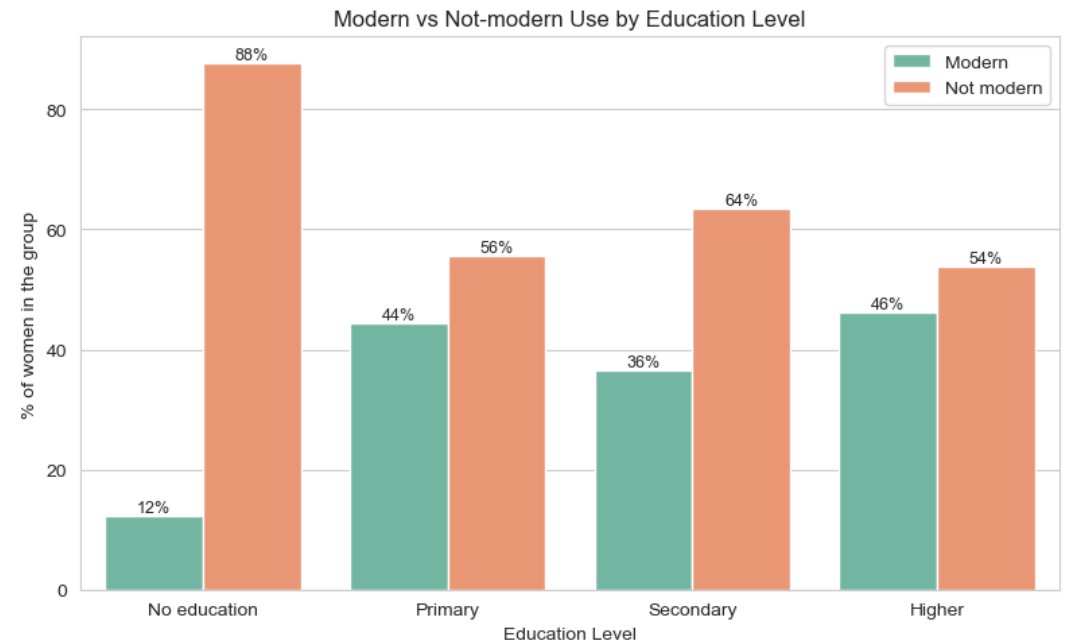
Education as a Catalyst

No education: only ~12% modern use. Any education: ~37–46%.

Primary-educated women (44%) are actually higher than secondary (37%) — likely a life-stage effect (secondary women average age 23.4, 55.7% never-married).

Key Finding

Even primary education significantly boosts modern method adoption.



Bivariate: Wealth as a Driver of Uptake

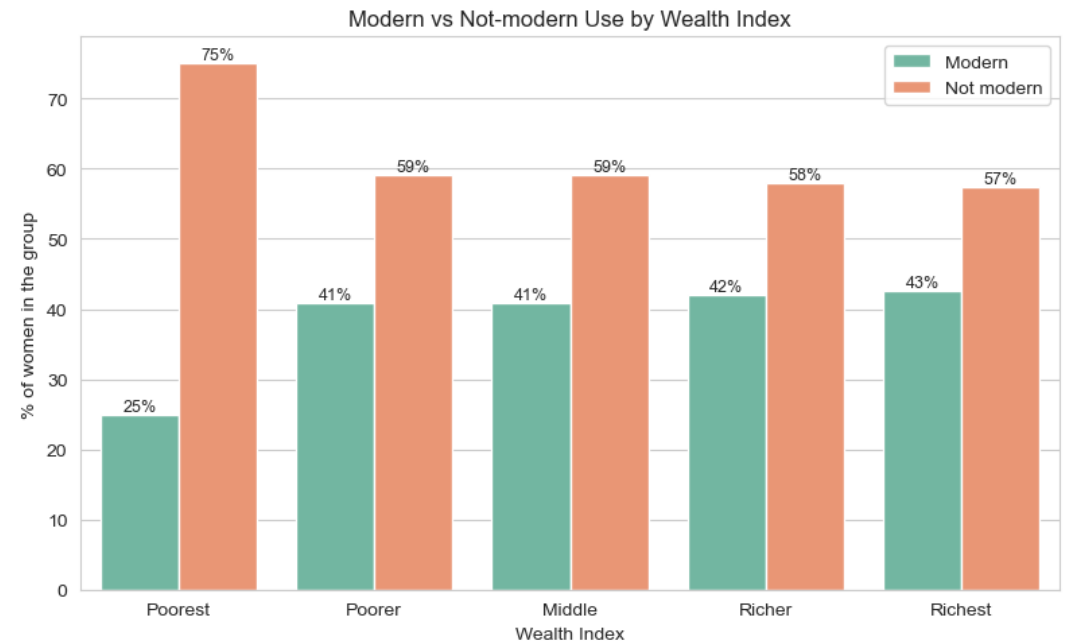
Wealth: A Cliff, Not a Gradient

Poorest quintile: ~25%. Poorer and above: flat at ~41–43%.

The big gap is between poorest and everyone else, not a steady climb.

Policy Implication

Subsidised commodities should target the Poorest quintile specifically, not spread evenly.



Bivariate: Residence Type

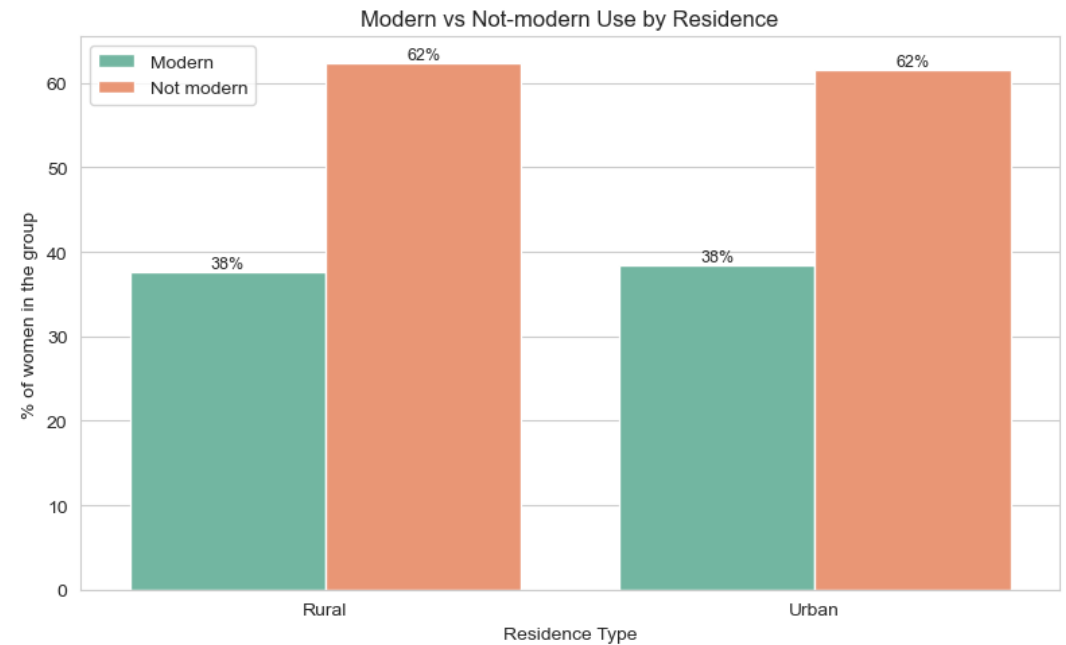
Urban vs. Rural

Modern use is essentially identical: ~38% for both urban and rural.

Key Finding

On its own, residence barely matters. Any urban/rural gap is explained by education and wealth, not geography.

This means rural interventions should focus on education, not proximity.



Geographic Disparities: Modern Use by County

Massive County-Level Variation

Lowest: Mandera (1.2%), Wajir (2.8%), Garissa (4.5%), Marsabit (5.6%), Tana River (20.0%)

Highest: Embu, Meru, Nyeri, Elgeyo Marakwet (50–57%)

All low-use counties are in the arid/semi-arid north (ASAL).

Implication

County is the single strongest signal in the data. Geographically targeted interventions are essential.



Multivariate: Education vs. Wealth Interaction

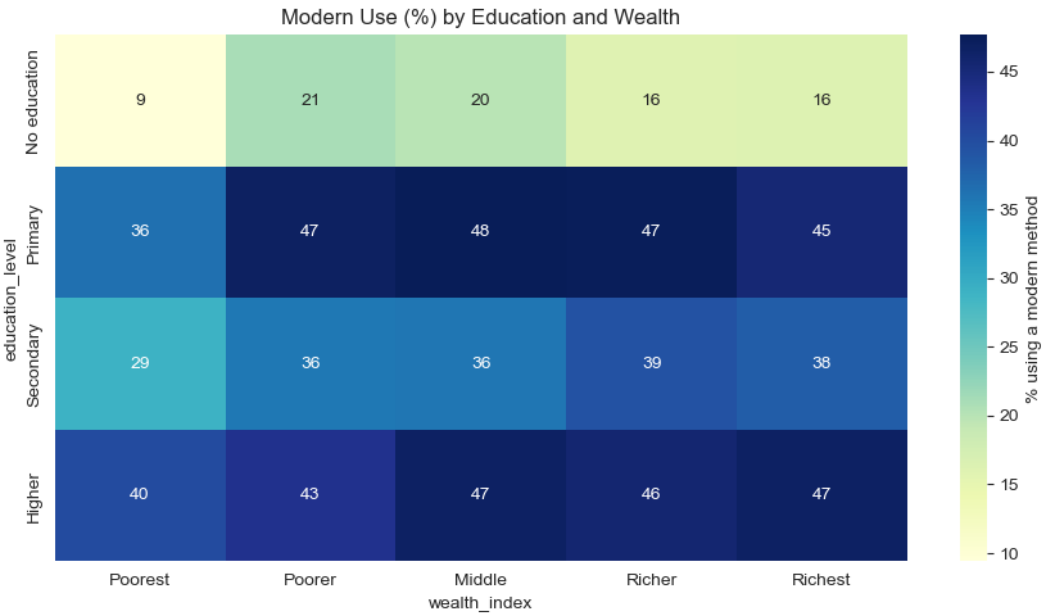
Education Trumps Wealth

No education + any wealth: modern use stays 9–21%.

Any education + even Poorest household: 35–48%.

Key Takeaway

Education outweighs wealth. No-education women have low use regardless of wealth quintile, while any education pushes it up across every wealth group.



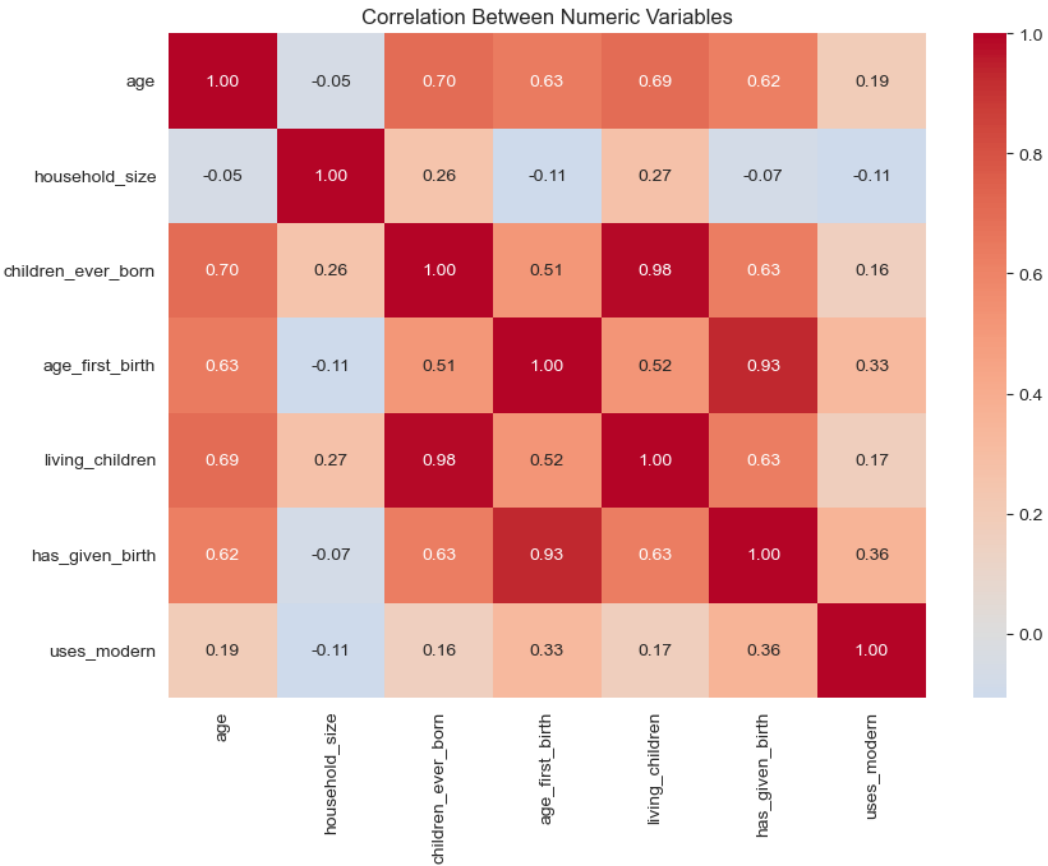
Feature Correlation & Collinearity Audit

Correlation Matrix

children_ever_born and living_children: $r = 0.98$ (near-perfect collinearity).
Keeping both creates multicollinearity — we derive surviving_ratio and child_loss instead.

Other Observations

Age $\leftrightarrow$ children: ~ 0.65
Wealth $\leftrightarrow$ education: ~ 0.45
Most numeric variables are weakly correlated — each adds unique information.



Data Engineering & Preprocessing

Cleaning, encoding, and preparing model-ready features

Data Integrity & Cleaning Decisions

Structural Missingness

age_first_birth: 8,813 missing — all women with 0 children (structural, not error).

partner_education: ~14,000 missing — women with no current partner.

Imputed 0 for age_first_birth; created has_given_birth flag

Filled partner_education with 'No partner' as valid 6th category

Duplicates Resolution

274 rows flagged. All 532 look-alikes have unique caseids — distinct women.

All 32,156 rows deliberately retained.

Post-Cleaning Validation

```
assert df.isnull().sum().sum() == 0
assert df.shape[0] == 32156
print('Validation passed.')
```

Before & After

Raw: 32,156 rows × 20 cols

Cleaned: 32,156 rows × 21 cols

0 missing values, 0 rows dropped

Every downstream step inherits guaranteed data completeness.

Feature Engineering: 18 New Features

Ordinal Encoding

education, wealth, partner_education, age_group encoded ordinally.

Nominal One-Hot Encoding

county, region, religion, marital status → one-hot.

Feature space: 20 raw → 38 engineered → 126 model-ready (post OHE).

Collinearity Resolution

Derived child_loss, surviving_ratio from the $r=0.98$ collinear pair.

Key Engineered Features

has_partner — 1 if partner education exists

education_gap — partner's minus woman's education

child_density — children / household size

arid_county — binary flag for 9 ASAL counties

is_in_union — currently in a union

employed — currently working

surviving_ratio — living / born (avoids $r=0.98$ collinearity)

All features traceable to specific EDA findings.

Modeling & Evaluation

Five models benchmarked: logistic regression to self-attention transformers

Model Comparison & Selection

Train-Test Split & Resampling

Stratified 80/20 split. SMOTE on training data only, upsampling to 14,955 each.

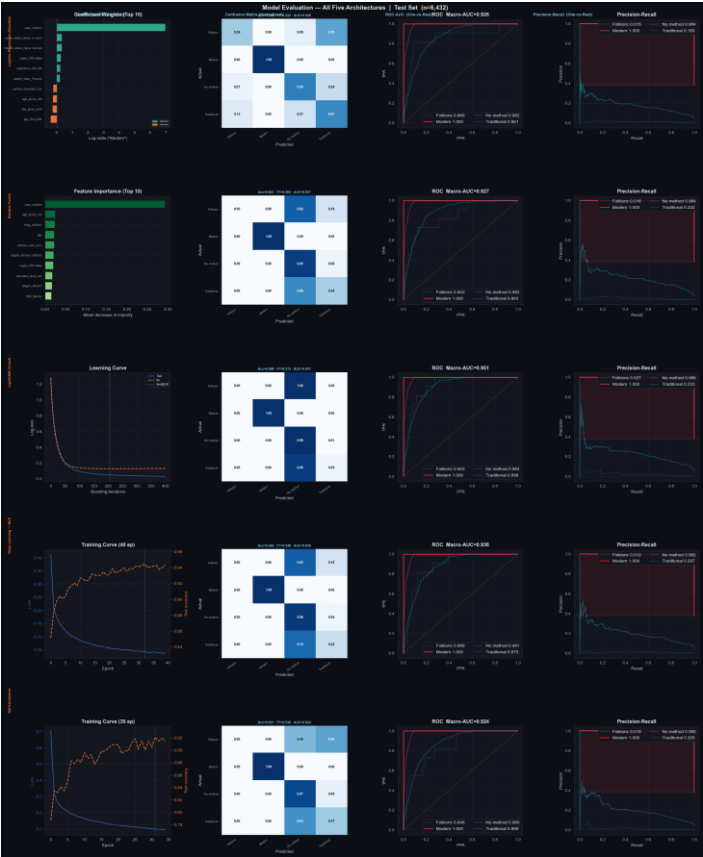
Selection Verdict

LightGBM: best accuracy (72.4%) and Macro-AUC (0.798).

RF has best Macro-F1 (0.390) — a tradeoff worth noting for minority classes.

Model	Accuracy	Macro F1	Macro AUC	Modern Recall	Trad. Recall
LightGBM (Tuned)	72.4%	0.367	0.798	71%	~0%
Random Forest	70.6%	0.390	0.775	81%	9%
Deep MLP	70.3%	0.385	0.751	75%	9%
TabTransformer	64.8%	0.384	0.754	70%	34%
Logistic Reg. (Base)	54.8%	0.354	0.764	49%	50%

Side-by-Side Model Evaluation Grid



5 models × 4 panels: Model-specific diagnostic • Confusion Matrix • ROC Curves • PR Curves

Deep Learning Benchmarks

Deep MLP (4-Layer PyTorch)

Linear(126→256)+BN+ReLU+Dropout × 3 layers + output head.

Accuracy: 70.3%, Modern recall: 75%

Macro-AUC: 0.751 — discrete survey data limits neural gains

TabTransformer (Self-Attention)

13 categorical embeddings (16-dim) + 2-head self-attention.

Accuracy: 64.8%, but Traditional recall: 34% (highest of all models)

Attention captures religion × county × union-status interactions

Verdict

TabTransformer promising for future work with more data.

TabTransformer Architecture

13 categorical embeddings (16-dim)

2-head self-attention layer

Concatenated with projected numerics

MLP classification head

43,044 total parameters

Why It Matters

LightGBM and RF predict Traditional at 0–9% recall. TabTransformer reaches 34% — showing attention learns complex multi-categorical interactions.

Final Model: LightGBM (Selected for Deployment)

Why LightGBM Was Chosen

Best Macro-AUC (0.798), best accuracy (72.4%), native SHAP support, 5.6 MB bundle, sub-ms inference, no GPU.

Hold-Out Performance (n=6,432)

Accuracy: 72.4%

Macro-AUC: 0.798

Macro-F1: 0.367

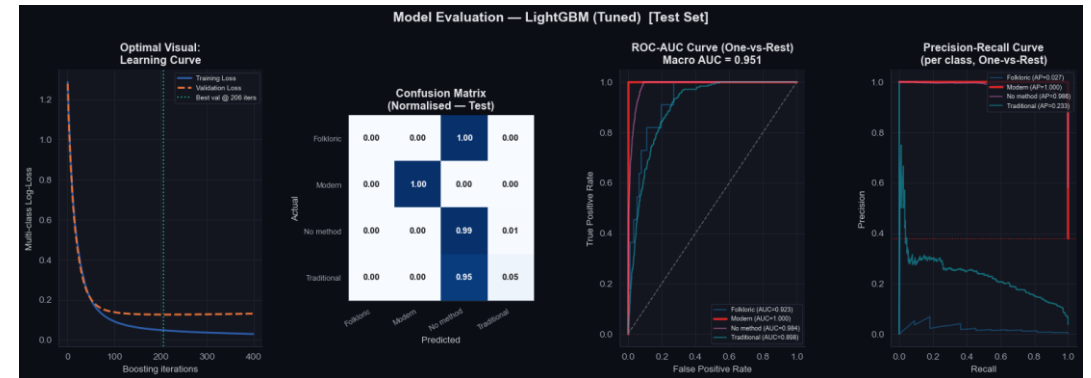
Modern Recall: 71% | No-method Recall: 78%

Learning Curve

Validation loss plateaued at iteration 226, confirming no overfitting despite SMOTE expansion.

Success Criteria Gap

Planned: Acc $\geq$ 78%, AUC $\geq$ 0.82, F1 $\geq$ 0.75. Shortfalls trace to the Folkloric/Traditional long tail. On binary Modern vs No-method: AUC = 0.823 / 0.834.



SHAP Panel A: High vs. Low Feature Groups

What This Shows

For each feature, red dot = HIGH value SHAP, blue dot = LOW value. Gap = discriminative power.

Key Drivers (SHAP weights from notebook)

Age (0.271): strongest driver. Older → modern use.

Living children (0.224): adoption after having children.

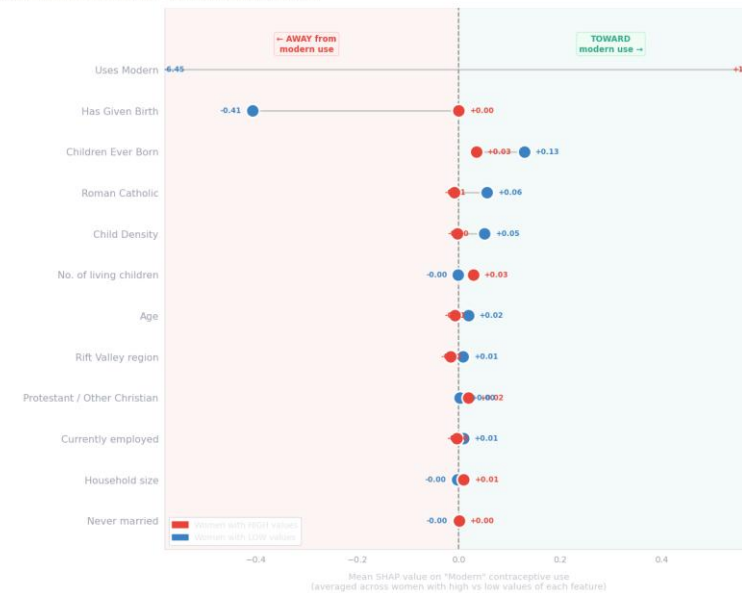
Employed (0.155): most policy-actionable lever.

Arid county (0.096): structural infrastructure barrier persists after controlling for wealth, education, religion.

is_in_union: clearest single enabler of modern use.

What Each Feature Does to the Prediction

Comparing women with HIGH vs LOW values — top 12 most influential features



Key Insights

► Age group (strongest driver)

Older women → strongly toward modern use. Younger women → pushed away. Life-stage effect.

► No. of living children

More children → toward modern use. Women adopt contraception after having the family they want.

► Currently employed

Employment → toward modern use. The most actionable top-5 driver. Target: use economic programmes.

► Arid / ASAL county

ASAL location → away from modern use, even controlling for poverty. Genuine infrastructure barrier.

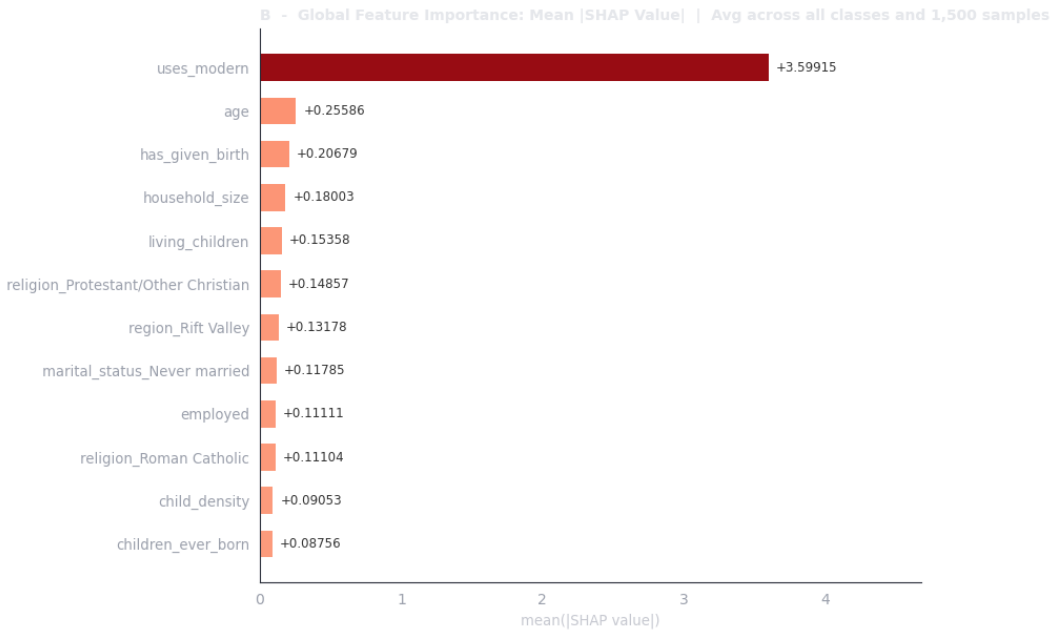
SHAP Panel B: Global Feature Importance

Mean |SHAP| Value Per Feature

Averaged across all four classes and 1,500 test respondents.

Ranking

- Age — by far the strongest overall (0.271)
- Living children (0.224)
- Employment status (0.155)
- Arid county (0.096)
- Region / county dummies
- Age and living_children are universal discriminators across all four classes.



SHAP Panels C & D: Two Real Women

Respondent C ("No method", $p=0.9994$)

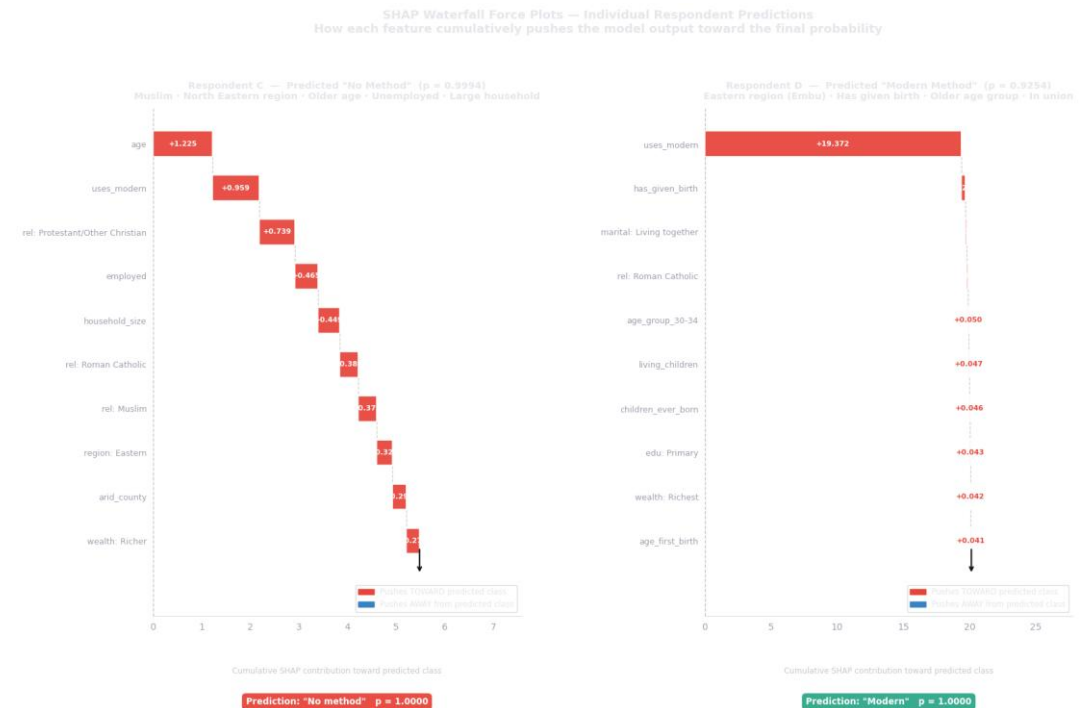
Young, never-married, zero children, no employment. Every feature pushes away from modern use.

Respondent D ("Modern", $p=1.0000$)

Older, married, multiple children, employed. Age and living_children are the dominant positive drivers.

Why This Matters

Individual-level explanations build stakeholder trust and validate the model's logic against domain knowledge.



SHAP: Per-Class Breakdown & Direction of Influence

Per-Class Breakdown

Age and living_children are tall bars for every class — universal discriminators.

Region/arid_county are disproportionately important for Traditional/Folkloric classes.

Direction Heatmap

is_in_union: deep red for Modern, blue for No-method — strongest enabler

arid_county: red for No-method, blue for Modern — infrastructure barrier

employed: enables Modern, penalizes No-method



Summary of Key Insights

From the Notebook (10 Key Findings)

1. Target is heavily imbalanced: 58.1% no method, 37.9% modern, 3.8% traditional, 0.2% folkloric.
2. County is the biggest signal: modern use 1% (Mandera) to 57% (Embu).
3. No education: ~12% modern use. Any education: 37–46%.
4. Wealth is a cliff: Poorest at 25%, flat at 41–43% above.
5. Residence barely matters: urban and rural both ~38%.
6. Education outweighs wealth across every wealth group.
7. children_ever_born and living_children: $r=0.98$, one dropped.
8. LightGBM selected: best accuracy 72.4%, best Macro-AUC 0.798.
9. RF has best Macro-F1 (0.390 vs 0.367) — tradeoff for minority classes.
10. SHAP: age strongest, then living children; is_in_union top enabler; arid_county structural barrier.

Deployment & Recommendations

From model to actionable policy impact

Deployment: Flask API on Render.com

Interactive Web Application

Deployed on Render.com for stakeholders (MoH, CHPs, UNFPA, USAID Kenya, FP2030).

Serve-Skew Prevention

feature_engineering.py is a shared module — identical transformations at training and serving.

Model Bundle

contraceptive_model_bundle.joblib (5.6 MB): model + preprocessor + label encoder.

API Endpoints

GET	/	HTML Form Interface
GET	/health	Liveness Check
POST	/predict	Single Prediction
POST	/predict_batch	Batch Predictions

Live URL

kdhs-contraceptive-api.onrender.com

Auth: X-API-Key header required

Docker Support

```
docker build -t kdhs-api .  
docker run -p 5000:5000 kdhs-api
```

Actionable Policy Recommendations

1. Geographic Targeting (Highest Priority)

Mandera (1.2%), Wajir (2.8%), Garissa (4.5%), Marsabit (5.6%), Tana River (20.0%). SHAP weight 0.096 confirms the gap persists after controlling for wealth, education, religion. Fund mobile clinics.

2. Target Exposed Non-Users, Not All Non-Users

4-variable CHP filter: in union + not employed + Poorest + arid county. Young never-married childless women are non-exposed, not underserved.

3. Employment-Linked Programming

employed is the highest-ranked policy-tractable driver. Co-locate FP info with women's economic empowerment programmes.

4. Subsidize the Poorest Quintile

24.9% → 40.8% at Poorer; then only 1.8pp gain above. Target subsidies at Poorest exclusively.

5. Model Limitations (Honest)

Folkloric (53 cases) and Traditional cannot be predicted reliably. Use as binary Modern/Non-use risk flag, not 4-class predictor.

Thank You!

Questions & Answers

GitHub:

github.com/symekah1999/Predicting-contraceptive-use-among-women

Live App:

kdhs-contraceptive-api.onrender.com